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Phonetically Grounded Neural Models for Cross-Lingual Historical Toponym Matching

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Abstract

Historical toponym resolution remains challenging when orthographic variation, script differences, and transliteration ambiguity obscure phonetic continuity. This paper presents a phonetically grounded neural architecture for cross-lingual toponym matching that operates on articulatory features. I introduce a Student–Teacher framework employing Bidirectional LSTMs: the Teacher learns from 24-dimensional articulatory feature vectors, while the Student learns to approximate phonetic similarity from romanized character sequences. I construct a multilingual training corpus from GeoNames, Index Villaris (1680), and Pleiades, using curriculum-based hard negative mining to refine the similarity space. Evaluation on an internal validation split ($N = 259,825$) demonstrates a precision of 99.6% at a recall of 98.5%. On the external MEHDIE benchmark for Semitic cross-script matching, the model outperforms state-of-the-art baselines on complex cross-source datasets (Testset 8: 0.74 vs. 0.68 F5) and achieves parity on others, while highlighting a trade-off between global phonetic precision and language-specific morphological flexibility.

Preprint notice. This manuscript is a preprint.

1 Introduction

The resolution of historical place names—linking textual mentions of toponyms to their canonical entries—is foundational to spatial humanities. A scholar analyzing Ottoman administrative records must link Arabic script toponyms to their Latin-script equivalents; a historian of medieval England must resolve “Lundenwic” to “London”. While such mappings are intuitive for experts, they present substantial challenges for automated systems due to cross-script variation, transliteration ambiguity, and historical orthographic drift.

This work addresses the *name-matching* component of resolution. I hypothesize that beneath orthographic diversity lies a unifying principle: toponyms possess phonetic identities (what I term *topophones*) that persist across scripts and time. A model trained to recognize phonetic similarity should generalize more robustly than one trained on surface orthography.

This paper makes the following contributions:

- A **phonetically-grounded neural architecture** using articulatory feature vectors derived from the International Phonetic Alphabet (IPA).
- A **Student-Teacher framework** enabling efficient inference on character sequences without runtime G2P conversion.
- A **large-scale training corpus** combining contemporary (GeoNames) and historical (Index Villaris, Pleiades) data.
- **Empirical evaluation** demonstrating state-of-the-art performance on difficult cross-source benchmarks while maintaining high precision for general search tasks.

The remainder of this paper is organized as follows. Section 2 reviews related work on toponym resolution, phonetic similarity, and neural approaches to string matching. Section 3 describes the model architecture in detail. Section 4 presents the construction of the training corpus. Section 5 outlines the experimental methodology. Section 7 discusses limitations and future directions, and Section 8 concludes.

2 Related Work

2.1 Toponym Resolution and Geocoding

Toponym resolution encompasses two subtasks: recognition (identifying place name mentions in text) and disambiguation (linking mentions to gazetteer en-

tries). Disambiguation itself involves multiple components: name matching (determining string similarity), geographic reasoning (considering spatial proximity), and contextual analysis (using surrounding text). This work addresses the *name matching* component specifically, learning phonetic similarity between strings. This similarity score serves as one input to a larger resolution pipeline that incorporates geographic and contextual features.

Classical approaches employ string similarity metrics. Recchia and Louwerse (Recchia and Louwerse, 2013) evaluated 21 similarity measures on romanized toponyms from 11 countries, finding that optimal measures varied by language but were consistent within linguistic families. Santos et al. (Santos et al., 2018) demonstrated that combining multiple string metrics with machine learning classifiers improved matching accuracy for Portuguese toponyms.

Rule-based phonetic algorithms like Soundex (Russell, 1918) and Metaphone (Philips, 1990) encode names into canonical phonetic representations, enabling matching of homophones. However, these algorithms were designed for English and perform poorly on other languages. The Double Metaphone algorithm (Philips, 2000) extended coverage to European languages but remains limited for non-Indo-European languages and cross-script scenarios.

2.2 Cross-Lingual and Historical Toponym Matching

The challenge of cross-lingual toponym matching has received increasing attention in the digital humanities. Coll Ardanuy and Sporleder (Coll Ardanuy and Sporleder, 2017) combine geographic and semantic features for historical toponym disambiguation in Dutch and German newspaper corpora, achieving strong results by exploiting Wikipedia-derived context and spatial clustering. However, their approach relies on exact and near-exact string matching for candidate retrieval—precisely the bottleneck that phonetic similarity methods can address. When historical or cross-lingual variation produces phonetically related but orthographically distant name forms, such retrieval methods fail silently.

Most relevant to this work is the MEHDIE project (Sagi et al., 2025), which developed tools for matching toponyms between historical Hebrew and Arabic sources in the Middle East. Sagi et al. present a benchmark comprising place pairs from 13th-century Arabic geographical texts (Yāqūt al-Ḥamawī’s *Kitāb Mu’jam al-Buldān*) and 12th-century Hebrew travel narratives (Benjamin of Tudela), along with the al-Turayyā and Kima gazetteers. Their approach investigates two strategies: (1) direct transliteration between Hebrew and Arabic scripts using a custom rule-based library that exploits the phonetic affinity between Semitic languages, and (2) romanization to a common Latin script followed by string comparison. They compare multiple string similarity metrics (Levenshtein, Jaro, Jaccard) and

phonetic distance measures, finding that direct transliteration between related scripts often outperforms romanization due to the loss of phonetic information in standard Latin transliteration schemes.

The MEHDIE work provides crucial insights for this approach. First, it demonstrates that phonetic relationships between languages can be exploited for toponym matching, validating the central hypothesis. Second, it highlights the limitations of transliteration-based approaches: their custom transliteration library required substantial linguistic expertise to develop and remains specific to Hebrew-Arabic pairs. Third, their benchmark reveals that no single metric dominates across all dataset pairs, suggesting that learned representations may offer advantages over hand-crafted similarity functions. Notably, the MEHDIE paper does not engage with recent deep learning approaches to toponym matching, focusing instead on interpretable rule-based and metric-based methods. The neural architecture presented here complements their work by learning phonetic similarity patterns from data rather than encoding them manually.

Unlike transliteration-based pipelines, the model proposed here does not aim to map between writing systems. Instead, it learns a shared similarity space grounded in articulatory properties, enabling recognition of phonetic correspondence without language-pair-specific rules.

There are clear synergies between the MEHDIE project and this work: MEHDIE’s matching software was originally based on an earlier version of the World Historical Gazetteer, and I have benefited from discussions with Tomer Sagi about cross-script toponym matching challenges. The present work can be seen as extending the WHG infrastructure that MEHDIE built upon, now incorporating learned phonetic representations to complement the rule-based approaches they developed.

The HIPE shared tasks (Ehrmann et al., 2020, 2022) established benchmarks for named entity recognition and linking in multilingual historical newspapers, highlighting the challenges posed by OCR errors, historical spelling variation, and cross-lingual entity references. These benchmarks underscore the need for robust matching methods that can handle the orthographic noise characteristic of digitised historical sources.

2.3 Deep Learning for Toponym Matching

Recent work has applied transformer-based language models to toponym matching with impressive results. Qiu et al. (Qiu et al., 2024) proposed GeoBERTTM, combining a BERT model pre-trained on Chinese geographic text (GeoBERT) with the Enhanced Sequential Inference Model (ESIM) architecture for toponym pair classification. Their approach integrates multiple Chinese-specific features including Pinyin romanization, Wubi keyboard encoding, character radicals, and stroke

counts. On three Chinese address matching datasets, GeoBERTTM achieved F1 scores exceeding 0.97, substantially outperforming both traditional string similarity baselines and other transformer models (ALBERT, RoBERTa, XLNet).

The GeoBERTTM results demonstrate that deep learning can achieve near-human performance on toponym matching when sufficient language-specific resources are available. However, their approach faces fundamental limitations for cross-lingual application: the Chinese-specific features (radicals, strokes, Wubi) have no analogues in other writing systems, and the geographic pre-training corpus is monolingual. Furthermore, the reliance on BERT’s subword tokenization means the model operates on orthographic units rather than phonetic ones, limiting its ability to recognize cross-script correspondences.

This work differs from GeoBERTTM in three key respects. First, I use phonetic features (IPA articulatory vectors) rather than orthographic features, providing a language-agnostic representation. Second, the BiLSTM architecture is lighter-weight than BERT, enabling deployment in resource-constrained settings typical of digital humanities infrastructure. Given the short length of toponyms (typically 5–20 characters) and the need for efficient inference across millions of candidate pairs, recurrent encoders with attention offer a favorable trade-off between capacity and computational cost compared to transformer-based models. Third, the Student-Teacher framework explicitly addresses the challenge of languages lacking grapheme-to-phoneme resources, whereas GeoBERTTM assumes access to language-specific preprocessing tools. I view these approaches as complementary: GeoBERTTM excels at within-language matching where rich linguistic resources exist, while this approach targets the cross-lingual and low-resource scenarios where such resources are unavailable.

2.4 Phonetic Representations in NLP

The use of phonetic features in natural language processing has a long history. Epitran (Mortensen et al., 2018) provides rule-based grapheme-to-phoneme conversion for over 100 languages, outputting IPA transcriptions. PanPhon (Mortensen et al., 2016) maps IPA segments to articulatory feature vectors encoding phonological properties (e.g., [+nasal], [-voiced], [+labial]).

Rama (Rama, 2016) applied Siamese convolutional networks to cognate identification using character embeddings, demonstrating that neural architectures can learn sound correspondences between related languages. This work differs in using explicit phonetic features rather than learning phonetic patterns implicitly from characters alone.

Recent work on multilingual named entity transliteration (Benites et al., 2020) has produced large-scale resources like TRANSLIT containing 1.6 million name

variants across 180 languages. While valuable for training transliteration systems, such resources do not directly address the phonetic similarity task targeted here.

2.5 Siamese Networks for Similarity Learning

Siamese networks (Bromley et al., 1993; Chopra et al., 2005) learn similarity functions by processing two inputs through identical sub-networks and comparing their output representations. They have been successfully applied to face verification (Schroff et al., 2015), signature verification (Bromley et al., 1993), and text similarity (Mueller and Thyagarajan, 2016; Neculoiu et al., 2016).

For text matching, Mueller and Thyagarajan (Mueller and Thyagarajan, 2016) demonstrated that Siamese LSTMs with Manhattan distance achieve strong performance on sentence similarity tasks. Neculoiu et al. (Neculoiu et al., 2016) applied character-level Siamese LSTMs to job title matching, showing that the architecture can learn both semantic and structural similarity.

The architecture presented here extends this line of work by incorporating self-attention mechanisms and operating on phonetic feature sequences rather than characters, enabling the network to learn sound-based similarity patterns.

3 Model Architecture

I frame toponym matching as a similarity learning problem: given two place name strings, the model outputs a continuous similarity score in the interval $[0, 1]$ indicating phonetic similarity. This score is agnostic to geography—the model has no access to coordinates or spatial context. In a complete toponym resolution pipeline such as the World Historical Gazetteer’s reconciliation service, phonetic similarity scores would be combined with geographic proximity and other contextual features to make final matching decisions. The contribution here is the phonetic similarity component alone.

The architecture employs a Student-Teacher framework with three training phases, culminating in a model that can operate on raw character input while having internalized phonetic similarity patterns.

3.1 Overview

The complete system comprises two parallel encoding pathways:

1. **Teacher (Phonetic) Encoder:** Processes IPA feature sequences derived from grapheme-to-phoneme conversion. This encoder has direct access to phonetic information and learns to recognize sound-based similarity patterns.

2. **Student (Character) Encoder:** Processes romanized character sequences with language identification. Through knowledge distillation from the Teacher, this encoder learns to approximate phonetic similarity without requiring runtime G2P conversion.

Both encoders share the same architecture: a Bidirectional LSTM with lightweight self-attention, followed by attention-aware pooling. The key difference lies in their input representations. Figure 1 illustrates the complete architecture.

3.2 Input Representations

3.2.1 Phonetic Features (Teacher)

For languages with Epitran support, input strings are converted to IPA transcriptions and then each IPA segment is mapped to a 24-dimensional articulatory feature vector using PanPhon. These binary features encode phonological properties including:

- Manner of articulation: $[\pm\text{syllabic}]$, $[\pm\text{consonantal}]$, $[\pm\text{sonorant}]$, $[\pm\text{continuant}]$, etc.
- Place of articulation: $[\pm\text{labial}]$, $[\pm\text{coronal}]$, $[\pm\text{dorsal}]$, $[\pm\text{pharyngeal}]$, etc.
- Voicing and secondary articulations: $[\pm\text{voice}]$, $[\pm\text{nasal}]$, $[\pm\text{lateral}]$, $[\pm\text{delayed release}]$, etc.

A toponym is thus represented as a sequence of feature vectors $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]$ where each $\mathbf{x}_t \in \{-1, 0, +1\}^{24}$ encodes the articulatory configuration of the t -th phoneme.

This representation offers several advantages over character-based approaches:

- Phonetically similar sounds (e.g., $[\text{p}]$ and $[\text{b}]$, differing only in voicing) have similar feature vectors.
- The representation is language-agnostic: the same features describe sounds across all human languages.
- Cross-script correspondences are captured naturally: “Moscow” $[\text{'m}\text{ɒ}\text{s}\text{k}\text{ə}\text{ʊ}]$ and “Москва” $[\text{m}\text{ɐ}'\text{s}\text{k}\text{v}\text{a}]$ produce similar feature sequences despite sharing no characters.

3.2.2 Character Embeddings with Language ID (Student)

For the Student encoder, AnyAscii¹ romanization is first applied to convert non-Latin scripts to Latin characters. AnyAscii provides ASCII transliteration for approximately 124,000 of Unicode’s 155,000 characters, covering all major scripts used in contemporary and historical toponymy.

A key property of AnyAscii is that it performs context-free, character-by-character transliteration. While this means romanization may not always match linguistically ideal conventions (e.g., context-dependent rules in Cyrillic languages), it ensures *consistency*: the same character always maps to the same ASCII output. This consistency is valuable for the Student network, which learns to recognize patterns in the romanized output rather than requiring linguistically perfect transliteration.

Each romanized character is represented as a learned embedding. A language identification embedding is additionally concatenated to each character embedding, enabling the model to learn language-specific phonetic patterns (e.g., that German “w” represents /v/ while English “w” represents /w/).

3.3 Sequence Encoder

Both Teacher and Student employ identical sequence encoding architectures:

3.3.1 Bidirectional LSTM

Input sequences are processed with a Bidirectional LSTM (Schuster and Paliwal, 1997) to capture context in both directions:

$$\vec{\mathbf{h}}_t = \text{LSTM}_{\text{fwd}}(\mathbf{x}_t, \vec{\mathbf{h}}_{t-1}) \quad (1)$$

$$\overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\text{bwd}}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}) \quad (2)$$

$$\mathbf{h}_t = [\vec{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{2d_h} \quad (3)$$

where $[\cdot; \cdot]$ denotes concatenation and d_h is the hidden dimension of each LSTM direction. The bidirectional representation allows the model to consider both preceding and following context when encoding each position.

3.3.2 Additive Self-Attention

Additive self-attention (Lin et al., 2017) is applied over the BiLSTM outputs to enable the model to focus on salient portions of the name. For toponym matching,

¹<https://github.com/anyascii/anyascii>

this is particularly valuable: root morphemes often carry more discriminative information than affixes (e.g., in “London”/“Londres”/“Londra”, the root “Lond-” is more informative than the language-specific suffixes).

Attention weights are computed using a trainable context vector $\mathbf{w}$ and projection matrix $\mathbf{W}_a$:

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_t))}{\sum_{t'} \exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_{t'}))} \quad (4)$$

The final embedding is computed as the weighted sum of hidden states:

$$\mathbf{e} = \sum_t \alpha_t \mathbf{h}_t \quad (5)$$

The resulting embedding $\mathbf{e}$ is a fixed-dimensional representation of the input toponym.

3.4 Similarity Computation

Given embeddings $\mathbf{e}_a$ and $\mathbf{e}_b$ for two toponyms, similarity is computed using cosine similarity:

$$\text{sim}(\mathbf{e}_a, \mathbf{e}_b) = \frac{\mathbf{e}_a \cdot \mathbf{e}_b}{\|\mathbf{e}_a\| \|\mathbf{e}_b\|} \quad (6)$$

Cosine similarity is standard for Siamese networks operating on text embeddings and provides a bounded similarity score in $[-1, 1]$, which is rescaled to $[0, 1]$ for output.

3.5 Three-Phase Training

Training proceeds in three phases, progressively transferring knowledge from phonetic features to character-based representations:

3.5.1 Phase 1: Teacher Pre-training

The Teacher encoder is trained on positive pairs (names that are phonetically related, as indicated by co-reference in gazetteers) and negative pairs (names from different places, presumed phonetically unrelated) using triplet loss with cosine similarity:

$$\mathcal{L}_{\text{triplet}} = \max(0, m - \text{sim}(\mathbf{e}_a, \mathbf{e}_p) + \text{sim}(\mathbf{e}_a, \mathbf{e}_n)) \quad (7)$$

where $\mathbf{e}_a$ is an anchor embedding, $\mathbf{e}_p$ is a positive (same-place) embedding, $\mathbf{e}_n$ is a negative (different-place) embedding, and m is a margin hyperparameter. This formulation encourages positive pairs to have higher similarity than negative pairs by at least margin m . This phase establishes the phonetic similarity manifold.

3.5.2 Phase 2: Student-Teacher Alignment

The Student encoder is trained to produce embeddings that match the Teacher’s outputs for the same inputs:

$$\mathcal{L}_{\text{align}} = \text{MSE}(\mathbf{e}_{\text{student}}, \mathbf{e}_{\text{teacher}}) + \lambda(1 - \cos(\mathbf{e}_{\text{student}}, \mathbf{e}_{\text{teacher}})) \quad (8)$$

The Teacher weights are frozen during this phase. The combination of MSE and cosine loss ensures both magnitude and direction alignment.

3.5.3 Phase 3: Student Fine-tuning with Curriculum Hard Negatives

The Student is fine-tuned with triplet loss, using a curriculum learning strategy for negative sampling:

- **Stage A:** Hard negatives are sampled based on orthographic similarity: pairs that are similar in character sequence but phonetically distinct (e.g., “Reading” the English town vs. “Reading” as a gerund).
- **Stage B** (optional): The Stage A model is used to mine false positives from the training set—pairs the model incorrectly judges as similar—which become hard negatives for further training.

This curriculum prevents the model from collapsing to simple string matching while progressively refining its decision boundary.

4 Dataset Construction

4.1 Source Data

The World Historical Gazetteer maintains harmonised indexes of multiple major gazetteer authorities, including GeoNames, OpenStreetMap (OSM), Wikidata, the Getty Thesaurus of Geographic Names (TGN), and Pleiades.² For training corpus construction, three sources were selected: GeoNames, Index Villaris, and Pleiades.

²<https://pleiades.stoa.org/>

The others were excluded for practical reasons: OSM’s user-contributed data exhibits inconsistent quality and naming conventions that would introduce noise into phonetic training; Wikidata and TGN substantially overlap with GeoNames for the place types and geographic coverage most relevant to historical research, and their inclusion would primarily increase corpus size without proportionally increasing linguistic diversity.

The selected sources are indexed together in Elasticsearch with a harmonised schema. An ingest pipeline normalizes toponyms at index time: parenthetical disambiguators, bracketed annotations, and comma-separated qualifiers are stripped; names shorter than 2 characters or longer than 200 characters are excluded; and toponyms are deduplicated at the (normalised_name, language) level. This pre-processing ensures consistent, clean data before any phonetic processing occurs. Hereafter, I refer to this combined, normalised dataset as the **WHG training corpus**.

4.1.1 GeoNames

GeoNames³ is a geographic database containing over 11 million place names with multilingual alternate names. Each place entry includes:

- A primary name and feature class (e.g., populated place, administrative division)
- Latitude/longitude coordinates
- Country and administrative codes
- Alternate names in multiple languages, tagged with ISO 639 language codes

Critically, all alternate names for a GeoNames entry refer to the *same geographic entity*. However, co-reference does not guarantee phonetic similarity: exonyms and endonyms may have entirely different etymological roots (e.g., “Germany”/“Deutschland”, “Japan”/“Nippon”, “Florence”/“Firenze”). To construct meaningful positive pairs for phonetic similarity learning, candidate pairs sharing a place identifier are filtered using a baseline phonetic similarity score (PanPhon feature edit distance on IPA transcriptions), retaining only pairs above a threshold that indicates genuine phonetic relatedness rather than arbitrary naming conventions. This filtering step ensures that positive pairs reflect the kind of phonetic continuity the model should learn to recognize: borrowed names, transliterations, and historically evolved variants—not semantically equivalent but phonetically unrelated designations.

³<https://www.geonames.org/>

4.1.2 Index Villaris (1680)

To incorporate historical name variants reflecting Early Modern English orthography, the training corpus includes *Index Villaris* (Gadd and Litvine, 2024), a digitised and georeferenced transcription of the cartographer John Adams’s 1680 gazetteer of England and Wales. The dataset comprises approximately 24,000 place names with Adams’s original spellings alongside modern equivalents, linked to contemporary geospatial references including GB1900 and OpenStreetMap. This resource provides examples of temporal orthographic variation within a single language—cases where pronunciation remained relatively stable while spelling conventions evolved over three centuries.

4.1.3 Pleiades

Pleiades⁴ is a community-built gazetteer of ancient places, providing coordinates, names, and attestations for locations in the Greek and Roman world and beyond. Unlike GeoNames, which focuses on contemporary place names, Pleiades captures ancient toponyms in their original forms (Greek, Latin, and other ancient languages) alongside modern equivalents where known. This provides valuable training signal for historical phonetic variation across millennia—for example, the evolution from Latin *Londinium* to modern *London*, or Greek *Ἀθῆναι* (Athēnai) to *Athens*.

4.2 Phonetic Annotation

Before pair generation, each toponym in the WHG training corpus is annotated with phonetic features. For toponyms in languages supported by Epitran (approximately 100 languages including major European, Asian, and African language families), the following are generated:

1. IPA transcription via Epitran grapheme-to-phoneme conversion
2. 24-dimensional articulatory feature vectors for each IPA segment via PanPhon

Toponyms where phonetic processing fails (e.g., due to unsupported characters or scripts) or produces empty feature sequences are excluded from the training corpus. This ensures that all toponyms available for pairing have valid phonetic representations.

⁴<https://pleiades.stoa.org/>

4.3 Pair Generation

Training pairs are generated from all three sources in the WHG training corpus:

4.3.1 Positive Pairs

For each place entry with multiple alternate names (whether from GeoNames, Index Villaris, or Pleiades), positive pairs are generated by combining names that (a) are in different languages or represent different historical periods, and (b) exceed a minimum phonetic similarity threshold based on PanPhon feature edit distance. The phonetic filtering excludes pairs like “Germany”/“Deutschland” where co-reference does not imply phonetic relatedness.

4.3.2 Negative Pairs

Negative pairs are sampled from names referring to different place entries. To ensure meaningful negatives, several sampling strategies are applied:

- **Random negatives:** Uniformly sampled from the corpus.
- **Geographic negatives:** Names from places in the same country or region, which may share linguistic patterns.
- **Orthographic negatives** (for curriculum learning): Names with high character-level similarity but different referents.

4.4 Corpus Statistics

Table 1: WHG training corpus statistics (preliminary)

Metric	Count
Total places processed	TBD
Unique toponyms with phonetic features	TBD
Positive pairs (phonetically valid)	TBD
Languages represented	TBD

4.5 Training Strategy: Balancing Contemporary and Historical Data

The three data sources differ dramatically in scale: GeoNames contributes millions of toponym pairs, while Index Villaris and Pleiades together yield only tens

of thousands. Naïve training on a combined corpus would result in historical examples constituting less than 1% of training batches, potentially causing the model to underfit the distinctive patterns of historical orthographic variation.

To address this class imbalance, I employ **stratified oversampling** during training. Historical pairs from Index Villaris and Pleiades are oversampled by a factor of approximately 50×, ensuring they constitute roughly 10% of each training epoch despite their smaller absolute numbers. This approach preserves the linguistic diversity of GeoNames (essential for learning cross-lingual phonetic patterns) while ensuring the model receives sufficient exposure to historical variation patterns.

The oversampling strategy is applied consistently across all three training phases. Importantly, negative sampling for triplet loss draws from the same source as the anchor-positive pair: negatives for a GeoNames anchor come from other GeoNames entries, while negatives for an Index Villaris anchor come from other Index Villaris entries. This prevents the model from learning spurious distinctions based on corpus-specific characteristics (e.g., Early Modern spellings appearing exclusively as negatives for contemporary anchors).

5 Experiments

5.1 Evaluation Benchmarks

5.1.1 Internal Validation Split (Generalization)

To measure the model’s robustness and precision on the training distribution (global toponyms), I evaluated on a held-out validation set of $N = 259,825$ pairs derived from the GeoNames/Pleiades corpus. This split evaluates the model as a “generalist” search engine.

Unlike external benchmarks, this evaluation directly reflects the intended deployment context: large-scale gazetteer reconciliation under extreme class imbalance, with millions of candidate pairs and high recall requirements.

5.1.2 MEHDIE Benchmark (Specialization)

I evaluate on the benchmark introduced by [Sagi et al. \(2025\)](#), comprising five dataset pairs from historical Hebrew and Arabic sources. This benchmark tests *cross-script* matching capabilities between Semitic languages, representing a severe domain shift from the largely Indo-European/Global training data.

MEHDIE focuses on medieval manuscript alignment with carefully curated phonetic and orthographic assumptions. Direct performance parity with MEHDIE

systems is therefore neither expected nor required for validation of the model’s core aims.

The MEHDIE datasets were not a design target of the proposed model; results are therefore interpreted as a stress-test of cross-corpus robustness rather than as a primary optimisation goal.

5.2 Implementation Details

The model was trained for 40 epochs on an NVIDIA A100 GPU. The curriculum learning strategy involved two mining phases: Stage A (orthographic negatives) and Stage B (model-based hard negatives). For Stage B, 180,873 hard negatives were mined using a hybrid strategy (70% random sampling, 30% targeted triplet scanning) to balance noise reduction with fine-grained discrimination. The final model (‘final_model_b.pt’) achieved a triplet loss of 0.0008.

6 Results

6.1 Internal Validation

The primary deployment context for this model is large-scale gazetteer reconciliation, where millions of candidate pairs must be evaluated efficiently with high precision. Table 2 presents performance on the held-out validation split ($N = 259,825$ pairs), comparing the trained model against string similarity baselines operating on the same AnyAscii-romanized input.

Table 2: Internal Validation Results ($N = 259,825$). The trained model substantially outperforms string similarity baselines, achieving higher recall at equivalent or better precision.

Method	θ	Precision	Recall	F1	F5
Levenshtein	0.50	0.999	0.883	0.937	0.887
	0.60	1.000	0.804	0.891	0.810
	0.70	1.000	0.719	0.837	0.727
Jaro-Winkler	0.60	0.942	0.956	0.949	0.955
	0.70	0.993	0.901	0.945	0.904
	0.80	0.999	0.804	0.891	0.810
This Model	0.60	0.984	0.994	0.989	0.993
	0.70	0.996	0.985	0.991	0.986
	0.80	1.000	0.958	0.978	0.962

At the optimal threshold ($\theta = 0.70$), the model achieves 99.6% precision with 98.5% recall, outperforming the best baseline (Jaro-Winkler at $\theta = 0.60$: F1=0.949) by over four percentage points. More significantly, the model maintains high recall at high precision thresholds: even at $\theta = 0.80$, recall remains above 95%, whereas Levenshtein at the same threshold recovers only 63% of true matches.

The learned embeddings achieve substantially better separation between positive and negative pairs than edit-distance metrics. Figure 2 illustrates this: 95% of negative pairs score below 0.485 with the trained model, compared to maximum negative similarities of 0.667 for Levenshtein and 0.900 for Jaro-Winkler. This separation enables confident automated matching at thresholds where baseline methods would produce unacceptable false positive rates.

6.2 MEHDIE Benchmark

The MEHDIE benchmark (Sagi et al., 2025) evaluates cross-script matching between historical Hebrew and Arabic sources—a severe domain shift from the predominantly Indo-European training data. Table 3 compares performance using the F-5 metric, which weights recall $5\times$ over precision to reflect the preference in historical research for high recall.

Table 3: MEHDIE Benchmark Results (Best F-5). MEHDIE baselines use linguistically engineered transliterations; AnyAscii baselines use naive character-by-character romanization matching the trained model’s input.

ID	Dataset Pair	MEHDIE Orth.	MEHDIE Phon.	AnyAscii Lev.	This Model	Δ Lev.
TS7	Yaqut (Ar) $\leftrightarrow$ Kima (He)	0.67	0.77	0.476	0.371	−0.105
TS8	Kima (He) $\leftrightarrow$ Thurayya (Ar)	0.65	0.68	0.750	0.738	−0.012
TS9	Tudela (He) $\leftrightarrow$ Thurayya (Ar)	0.92	0.70	0.768	0.676	−0.092
TS10	Yaqut $\leftrightarrow$ Kima (Maghreb)	0.75	0.77	0.428	0.437	+0.009
TS11	Damast (De) $\leftrightarrow$ Tudela (He)	0.78	0.88	0.832	0.857	+0.025
Average		0.75	0.76	0.651	0.616	−0.035

Four baselines are shown. The MEHDIE orthographic and phonetic baselines operate on carefully engineered transliterations designed for Hebrew-Arabic correspondence; these embed substantial linguistic expertise. The AnyAscii Levenshtein baseline uses the same naive romanization as the trained model, providing a fair comparison of what the learned representations add beyond simple edit distance.

The model outperforms AnyAscii Levenshtein on TS10 and TS11, and achieves near-parity on TS8 (0.738 vs. 0.750). On TS11 (German-Hebrew), the model achieves the second-best result overall (0.857), approaching the specialised MEHDIE phonetic baseline (0.88). This testset involves European-Semitic matching, where the

model’s global training data provides relevant signal.

Performance is weaker on TS7 and TS9, where the model underperforms even simple Levenshtein. Notably, however, the model achieves *higher recall* than Levenshtein on these testsets (TS9: 100% vs. 88.9%; TS7: 75.8% at $\theta = 0.9$ vs. 54.5% at $\theta = 0.7$), but at the cost of substantially more false positives. For applications prioritising recall over precision—such as candidate generation for human review—this trade-off may be acceptable.

The gap between MEHDIE’s engineered baselines and the AnyAscii baselines (average F-5: 0.76 vs. 0.65) quantifies the value of linguistic preprocessing for this domain. The learned model, operating on naive romanization, cannot fully compensate for this missing linguistic structure. However, on the internal validation benchmark (Table 2), where no such preprocessing advantage exists, the model substantially outperforms string similarity baselines, demonstrating that the learned representations capture genuine phonetic structure rather than merely approximating edit distance.

6.3 Deployment and Qualitative Evaluation

The phonetic similarity model is deployed as a core component of **WHG PLACE** (Place Linkage, Alignment, and Concordance Engine), the World Historical Gazetteer’s reconciliation infrastructure. Embeddings have been computed for all 68.7 million toponyms in its Elasticsearch index, enabling sub-second similarity search via kNN queries.

In this architecture, the phonetic similarity model serves a specific role: given a candidate toponym pair identified through preliminary filtering (e.g., geographic proximity, shared administrative region), the model produces a continuous similarity score that can be combined with other contextual features. The combined score informs both automated matching decisions and the ranking of candidates presented to human reviewers for validation.

This modular design allows the phonetic similarity component to be evaluated and improved independently while integrating seamlessly into the broader reconciliation pipeline. The model’s continuous output (rather than binary classification) is essential for this use case, as downstream components require graded similarity judgments to rank candidates and calibrate confidence thresholds.

To assess practical utility beyond aggregate metrics, I sampled random toponyms from two populations: (a) those with Epitran/PanPhon support (testing the full Teacher-Student pipeline), and (b) those without G2P resources (testing Student generalisation). For each query, the top 15 matches by cosine similarity were retrieved and manually evaluated for phonetic plausibility. Representative examples appear in [Appendix A](#).

7 Discussion

7.1 Internal Validation: The Core Result

The primary contribution of this work is demonstrated on the internal validation benchmark, where the model substantially outperforms string similarity baselines operating on identical input representations. At the optimal threshold, the model achieves $F1=0.991$ compared to Jaro-Winkler’s $F1=0.949$ and Levenshtein’s $F1=0.937$ —an improvement of 4–5 percentage points.

More significant than the aggregate metrics is the *separation* between positive and negative pairs. The learned embeddings push 95% of negative pairs below similarity 0.485, whereas Levenshtein negatives extend to 0.667 and Jaro-Winkler negatives reach 0.900. This separation has practical consequences: it enables confident automated matching at thresholds where baseline methods would produce unacceptable false positive rates. For large-scale gazetteer reconciliation involving millions of candidate pairs, this margin is essential.

7.2 The Generalist-Specialist Trade-off

The MEHDIE benchmark results reveal a fundamental tension in cross-lingual matching: models optimised for global consistency necessarily sacrifice sensitivity to language-family-specific patterns.

The MEHDIE system encodes explicit knowledge of Semitic consonantal roots, correctly treating radical vowel alternations (e.g., Arabic *kitāb* / Hebrew *ketav*) as identity-preserving. This model, trained on global data where vowel changes typically signal distinct entities (English *London* vs. *Linden*), learns a more conservative similarity function. This explains the performance gap on TS7 and TS9.

Notably, the model does not consistently outperform simple Levenshtein on MEHDIE testsets when both operate on AnyAscii romanization (average F-5: 0.616 vs. 0.651). However, this comparison is instructive: the model achieves *higher recall* on most testsets (e.g., TS9: 100% vs. 88.9%; TS11: 96.9% vs. 93.8%) at the cost of lower precision. For candidate generation workflows where human reviewers filter false positives, this trade-off may be preferable to the higher-precision, lower-recall profile of edit distance.

The gap between MEHDIE’s engineered baselines (average F-5: 0.76) and the AnyAscii baselines (0.65) quantifies the value of linguistic preprocessing for this domain. The learned model cannot fully compensate for missing linguistic structure in its input representation. This suggests that for specialised domains like Semitic cross-script matching, combining learned embeddings with linguistically informed preprocessing may yield the best results.

7.3 Embedding Space Characteristics

The 64-dimensional embedding space appears sufficient for the current task, but the 14% of positive pairs scoring below 0.9 similarity suggests headroom for improvement. These difficult cases warrant investigation: preliminary analysis indicates they cluster around (a) extreme historical drift (e.g., Latin to modern vernacular), (b) competing transliteration standards, and (c) borrowed names with phonological adaptation.

Increasing embedding dimensionality to 128 or 256 dimensions might capture more nuanced phonetic distinctions, but would increase computational cost both during training and at inference time when input strings must be encoded for search. A careful trade-off analysis would be required before deployment.

7.4 Limitations

7.4.1 Grapheme-to-Phoneme Coverage

Epitran supports approximately 100 languages, but historically significant languages (Middle Persian, Classical Tibetan, pre-modern Korean) lack resources. The Student encoder provides partial mitigation through transfer from related scripts, but truly out-of-distribution writing systems (cuneiform, hieroglyphics) would require dedicated G2P development or transliteration to supported scripts.

7.4.2 Pronunciation Variation

The model assumes canonical pronunciation, but toponyms often have multiple valid realisations (regional accents, historical shifts, spelling pronunciations). Future work could explore multi-embedding approaches or uncertainty quantification.

7.4.3 Phonetic vs. Geographic Disambiguation

The model cannot distinguish phonetically identical but geographically distinct places (Paris, Texas vs. Paris, France). Integration with geographic features in the WHG PLACE pipeline addresses this limitation at the system level.

7.4.4 Input Representation

The MEHDIE results demonstrate that naive romanization (AnyAscii) discards phonetic information that linguistically informed transliteration preserves. For cross-script matching between related languages (e.g., Hebrew-Arabic), the model’s

input representation may be a limiting factor independent of architectural capacity.

7.5 Future Directions

Several extensions merit investigation:

- **Semitic fine-tuning:** Incorporating MEHDIE benchmark data as curriculum training examples could improve performance on morphologically complex language pairs without sacrificing global precision.
- **Improved romanization:** Replacing AnyAscii with linguistically informed transliteration for specific script pairs could recover phonetic information currently lost in preprocessing.
- **Language and script identification:** Automatic detection of source language could improve Student encoder performance, particularly for scripts where the same character sequences have different phonetic realisations across languages.
- **Semantic encoding:** Classifying toponymic elements (e.g., “-bury”, “-chester”, “Saint-”) to capture geographic concepts could complement phonetic similarity with semantic features, though this remains at the conceptual stage.
- **Attention interpretability:** Visualising learned attention weights would likely reveal which phonetic segments carry discriminative information—one might expect consonant clusters and word-initial segments to receive higher weight, as these are typically most stable across transliteration systems.

8 Conclusion

I have presented a neural architecture for cross-lingual toponym matching that learns phonetic similarity from articulatory features. On the internal validation benchmark, the model substantially outperforms string similarity baselines (F1=0.991 vs. 0.949), with learned embeddings achieving markedly better separation between matching and non-matching pairs than edit-distance metrics.

On the external MEHDIE benchmark for Semitic cross-script matching, results are mixed: the model approaches or exceeds baseline performance on test-sets involving European-Semitic pairs (TS11: 0.857) but underperforms on Hebrew-Arabic pairs where specialised linguistic knowledge provides significant advan-

tage. This highlights a fundamental trade-off between global consistency and language-family-specific flexibility.

The model offers a scalable, language-agnostic approach to phonetic similarity that requires no expert rule-crafting and operates efficiently at scale. Embeddings for 68.7 million toponyms have been computed and deployed in the World Historical Gazetteer’s reconciliation infrastructure. The trained model and code are released to facilitate further research in global digital humanities.

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A Qualitative Evaluation Tables

Noise robustness: Latin script

Table 4: Noise robustness: **Kashkarivka** $\rightarrow$ **Kaashkarivka** [ga, latin]

Noise applied: duplication 'a', vowel 'a' > 'a'					
Recovery: original not in top-k					
Rank	Score	Name	Lang	Script	
Q	—	Kaashkarivka	ga	latin	
1	0.9699	Mechkarovtsi	en	latin	
2	0.9680	Acquafresca	fi	latin	
3	0.9653	Varo kyrka	fr	latin	
4	0.9648	Acquafresca	it	latin	
5	0.9641	Skrypskaya	und	latin	
6	0.9634	Acquafresca	pl	latin	
7	0.9624	Skrupski	und	latin	
8	0.9618	Bekkarvika	und	latin	
9	0.9614	Bekkarvika	no	latin	
10	0.9607	Asemakirkko	fi	latin	
11	0.9598	Bekkjarvika	und	latin	
12	0.9596	Bekkjarvika	no	latin	
13	0.9594	Acquafresca	ro	latin	
14	0.9589	Acquafresca	en	latin	
15	0.9586	Skorubica	und	latin	

Noise robustness: Cyrillic script

Noise robustness: Greek script

Noise robustness: Arabic script

Table 5: Noise robustness: **Dub V dobrem pive** → **Dub V dobem piev** [cs, latin]

Noise applied: transposition at 0, deletion 'r', transposition at 0, transposition at 14
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Dub V dobem piev	cs	latin
1	0.9810	Spruitenbosch	nl	latin
2	0.9772	Wedembosbeek	ga	latin
3	0.9772	Wedembosbeek	ceb	latin
4	0.9772	Wedembosbeek	und	latin
5	0.9738	Cotton Abbotts	nan	latin
6	0.9738	tombeau de Dubois	fr	latin
7	0.9736	Mutobafimbo	und	latin
8	0.9734	Mutobafimbo	ceb	latin
9	0.9734	Hochtannbergpass	sv	latin
10	0.9734	dunsbury farm	en	latin
11	0.9733	Cotton Abbotts	ceb	latin
12	0.9733	Cotton Abbotts	ga	latin
13	0.9732	Cotton Abbotts	lld	latin
14	0.9732	Cotton Abbotts	cy	latin
15	0.9732	Cotton Abbotts	und	latin

Table 6: Noise robustness: **Shaan-Kaya** → **Zhaan-Kaya** [uk, cyrillic]

Noise applied: consonant 'Sh'>'Zh'

Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Zhaan-Kaya	uk	cyrillic
1	0.9753	synagoga	sk	latin
2	0.9753	synagoga	und	latin
3	0.9743	sijangar	und	latin
4	0.9730	2004 Monaco GP	tr	latin
5	0.9693	can Roca	ca	latin
6	0.9662	1833NianSuMenDaLaDeZhen	zh	cjk
7	0.9661	shub nagar	und	latin
8	0.9658	can Moja	ca	latin
9	0.9629	con Bac Sa	vi	latin
10	0.9626	Cong hoa Ao	vi	latin
11	0.9620	ShunYiXinChengDi17JieQuSY00-0017-6006DeKuaiChuZhongYongDeXiangMu	und	cjk
12	0.9617	MiG-15VTZ No5	und	cyrillic
13	0.9615	can Pau Xic	ca	latin
14	0.9607	1679NianYaMeiNiYaDeZhen	zh	cjk
15	0.9600	1679NianSanHePingGuDeZhen	zh	cjk

Table 7: Noise robustness: **Horace M. Albright Training Center** → **Horace M. Albright Training Centerr** [ro, latin]

Noise applied: duplication 'r'				
Recovery: original not in top-k				
Rank	Score	Name	Lang	Script
Q	—	Horace M. Albright Training Centerr	ro	latin
1	0.9937	Fraccionamiento Metropolis II	en	latin
2	0.9909	Crow Meadow Preyterian Cemetery	und	latin
3	0.9904	British Guard Tower on HaYarkon Street	en	latin
4	0.9903	St Margaret of Scotland Cemetery	und	latin
5	0.9901	Stolperstein en memoria de Eugen Prager	es	latin
6	0.9896	Edward Frederick Primary School	und	latin
7	0.9896	Otdeleniye Sovkhoza Raduga Nomer Tri	und	latin
8	0.9894	Original Church of God Church Cemetery	en	latin
9	0.9894	Gordonsville Afro-American Cemetery	en	latin
10	0.9894	Rotorua South African War memorial	en	latin
11	0.9890	Borgund Lutheran Church Cemetery	und	latin
12	0.9889	Stolperstein en memoria de Hugo Meyer	es	latin
13	0.9887	Norton-on-Tees Queen Victoria Memorial	und	latin
14	0.9886	Protea Banks Marine Protected Area	en	latin
15	0.9886	Jaruga Hidroelektrik Santrali	tr	latin

Table 8: Noise robustness: **wrnsk** $\rightarrow$ **wrnsk** [fa, arabic]

Noise applied: transposition at 4

Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	wrnsk	fa	arabic
1	0.9891	Urnnsk	crh	latin
2	0.9888	Urnnsk	no	latin
3	0.9888	Urnnsk	war	latin
4	0.9885	Urnnsk	und	latin
5	0.9885	Zrnosek	nl	latin
6	0.9866	Zrnosek	fr	latin
7	0.9859	Fronsac	fr	latin
8	0.9846	Zrnosek	sr	latin
9	0.9828	Fronsac	sr	latin
10	0.9823	Fronsac	ms	latin
11	0.9820	Pronsk	nn	latin
12	0.9820	Pronsk	no	latin
13	0.9817	Pronsk	crh	latin
14	0.9816	Dornasac	ca	latin
15	0.9815	Pronsk	sl	latin

Table 9: Noise robustness: **Desa Bah Tonang** → **Dessaa Bah Tonang** [id, latin]

Noise applied: consonant 's'>'ss', duplication 'a'

Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Dessaa Bah Tonang	id	latin
1	0.9878	Bukit Batubandang	sv	latin
2	0.9872	Etang de Capeatang	de	latin
3	0.9860	Lathe of Sutton at Hone	en	latin
4	0.9855	Raftatangastong	und	latin
5	0.9844	Bukit Batubandang	nl	latin
6	0.9841	Goth Rabu Bhotani	und	latin
7	0.9838	Buxton Landing	und	latin
8	0.9838	okcidenta Lothian	eo	latin
9	0.9838	Pod Svatojankem	cs	latin
10	0.9837	Ruxton Landing	und	latin
11	0.9828	Etang de Capeatang	en	latin
12	0.9828	Opstina Huehuetan	sh	latin
13	0.9824	Koudiat Chatouane	ast	latin
14	0.9824	Koudiat Chatouane	und	latin
15	0.9823	Opstina Huehuetan	nl	latin

Table 10: Noise robustness: **Teatro Carlo Marengo** → **aetro Zarlo Narenoc**
[en, latin]

Noise applied: consonant 'M'>'N', transposition at 1, transposition at 18, deletion 'T', consonant 'C'>'Z'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	aetro Zarlo Narenoc	en	latin
1	0.9947	Monterrubio de la Serena	pt	latin
2	0.9943	Monterrubio de la Serena	pt	latin
3	0.9940	Zuccherificio Volano	nl	latin
4	0.9934	Szczawin Borowy-Kolonia	ast	latin
5	0.9934	Szczawin Borowy-Kolonia	und	latin
6	0.9932	Szczawin Borowy-Kolonia	nan	latin
7	0.9927	Zuccherificio Volano	sr	latin
8	0.9926	Zuccherificio Volano	it	latin
9	0.9920	Czerniejowa-Kolonii	genitive	latin
10	0.9917	Razvaliny Kir'yanovo	und	latin
11	0.9914	Rathaus von Verolengo	de	latin
12	0.9906	Zuccherificio Volano	tl	latin
13	0.9903	Yakun' Pereval'nyy	ru	latin
14	0.9902	Kjerrengvollfloen	no	latin
15	0.9897	Zuccherificio Volano	cs	latin

Table 11: Noise robustness: **Borne frontiere 253A** $\rightarrow$ **BBorn frontiere 253A**
[fr, latin]

Noise applied: duplication 'r', deletion 'e', duplication 'B'					
Recovery: original not in top-k					
Rank	Score	Name	Lang	Script	
Q	—	BBorn frontiere 253A	fr	latin	
1	0.9766	brownbread street	en	latin	
2	0.9728	borne frontiere LB 001	fr	latin	
3	0.9726	brouwerij Liefmans	en	latin	
4	0.9715	borne frontiere 101	fr	latin	
5	0.9668	Longbrooms Copse	en	latin	
6	0.9662	braniborska dieceze	cs	latin	
7	0.9652	Amborovoro Atsimo	und	latin	
8	0.9634	brown house bridge	en	latin	
9	0.9631	borne frontiere #179	und	latin	
10	0.9607	Springbokspoort	und	latin	
11	0.9605	proprieta Converso	it	latin	
12	0.9603	former Wagnerbrau inn	en	latin	
13	0.9596	Bruch von Gravenbruch	es	latin	
14	0.9591	Kvernbekkfallet	en	latin	
15	0.9590	Portone di bronzo	it	latin	

Table 12: Noise robustness: **Lyne** → **Lyne** [ga, latin]

Noise applied: vowel 'e' > 'e'

Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Lyne	ga	latin
1	0.9761	Al 'Ais	de	latin
2	0.9731	Plemy	af	latin
3	0.9731	Plemy	bar	latin
4	0.9731	Plemy	da	latin
5	0.9731	Plemy	eo	latin
6	0.9731	Plemy	fur	latin
7	0.9731	Plemy	ga	latin
8	0.9731	Plemy	is	latin
9	0.9731	Plemy	kab	latin
10	0.9731	Plemy	ku	latin
11	0.9731	Plemy	lb	latin
12	0.9731	Plemy	mul	latin
13	0.9731	Plemy	nan	latin
14	0.9731	Plemy	nn	latin
15	0.9731	Plemy	scn	latin

Table 13: Noise robustness: **Kul’turniy landshaft Bamians’koi dolini** → **Kur-tuurniy llandshaft Vamains’koi dooinli** [uk, cyrillic]

Noise applied: consonant ’l’>’r’, transposition at 34, duplication ’o’, deletion ’’, transposition at 22, transposition at 35,
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Kurtuurniy llandshaft Vamains’koi dooinli	uk	cyrillic
1	0.9915	station meteorologique d’Avignonet-Lauragais	fr	latin
2	0.9888	station meteorologique de Joinville	fr	latin
3	0.9882	station Colvecchio-Poggio Sommavilla	nl	latin
4	0.9881	stazione di Colvecchio-Poggio Sommavilla	it	latin
5	0.9880	station meteorologique de Montgivray	fr	latin
6	0.9877	santuario della Madonna del Patrocinio	it	latin
7	0.9872	station meteorologique de Magnanville	fr	latin
8	0.9869	station meteorologique de Saint-Gervais	fr	latin
9	0.9869	santuario della Madonna di Romvello	it	latin
10	0.9862	First Unitarian Congregation of Toronto	en	latin
11	0.9857	station meteorologique de Sainville	fr	latin
12	0.9856	station meteorologique d’Aubigny-sur-Nere	fr	latin
13	0.9850	station meteorologique de Saint-Jean-de-Minervois	fr	latin
14	0.9849	station meteorologique de Guillonville	fr	latin
15	0.9847	station meteorologique de Saint-Verand	fr	latin

Table 14: Noise robustness: **Montedzh Mauntin** $\rightarrow$ **Mntedzh MMAuntin** [ru, cyrillic]

Noise applied: deletion 'o', duplication 'M'					
Recovery: original not in top-k					
Rank	Score	Name	Lang	Script	
Q	—	Mntedzh MMAuntin	ru	cyrillic	
1	0.9840	Sungai Kamundan	ga	latin	
2	0.9835	Sungai Kamundan	ceb	latin	
3	0.9833	Sungai Kamundan	und	latin	
4	0.9804	Ronco Campo Canneto	sr	latin	
5	0.9801	'Ounda Oummounto	nl	latin	
6	0.9800	Town of Hammonton	und	latin	
7	0.9798	Sanga do Tamandua	und	latin	
8	0.9797	Gembonganomwetan	en	latin	
9	0.9782	Sungai Kumandan	und	latin	
10	0.9775	Dongjiamaimaitun	en	latin	
11	0.9768	Town of Lemington	en	latin	
12	0.9768	'Ounda Oummounto	ceb	latin	
13	0.9768	Town of Remington	en	latin	
14	0.9766	Wainikumbandamu	en	latin	
15	0.9765	Amaga-inbintha	my	latin	

Table 15: Noise robustness: **Mata-di-San-Zhuan** $\rightarrow$ **Matoa-di-So-Zhuan** [kk, cyrillic]

Noise applied: duplication 'a', deletion 'n', vowel 'a'>'o', vowel 'a'>'o'					
Recovery: original not in top-k					
Rank	Score	Name	Lang	Script	
Q	—	Matoa-di-So-Zhuan	kk	cyrillic	
1	0.9936	Argoub ez Zitouna	ceb	latin	
2	0.9936	Argoub ez Zitouna	und	latin	
3	0.9897	Qizilqoga awdani	kk	latin	
4	0.9891	Oakfield Township	fr	latin	
5	0.9887	Parking Lot Median	und	latin	
6	0.9876	Kloster Bogdana	de	latin	
7	0.9855	Alto Cubugon	ceb	latin	
8	0.9853	Alto Cubugon	und	latin	
9	0.9848	Osiedle Pogodna	und	latin	
10	0.9843	Sarikiz Meydani	und	latin	
11	0.9842	Alto de San Jouquin	nl	latin	
12	0.9839	Robathsiedlung	und	latin	
13	0.9830	Ilha Butui Pequena	pt	latin	
14	0.9830	Picacho El Tuetano	ceb	latin	
15	0.9828	Corrego do Caetano	sv	latin	

Table 16: Noise robustness: **Levkavshchyna** → **Lekvaavchyna** [tt, cyrillic]

Noise applied: duplication 'a', transposition at 2, deletion 'sh'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Lekvaavchyna	tt	cyrillic
1	0.9876	lac Bouleau Jaune	fr	latin
2	0.9844	Melekuonis	sgs	latin
3	0.9839	lago Don Juan	it	latin
4	0.9834	Pelekuonis	sgs	latin
5	0.9828	Il'chenkovo	und	latin
6	0.9820	lac a l'Eau Jaune	de	latin
7	0.9801	l'Agulla Fina	und	latin
8	0.9800	Ollukansuo	und	latin
9	0.9800	lac Elie-Gagnon	fr	latin
10	0.9799	lago Te Anau	vec	latin
11	0.9793	Cerro El Banqueo	es	latin
12	0.9786	Boca la Zanja	es	latin
13	0.9786	Kolikanaho	fi	latin
14	0.9784	laguna Yvera	und	latin
15	0.9778	lac Helene-Mann	fr	latin

Table 17: Noise robustness: **Urochyshcha Shnury** → **Yuroychshcha Shnuuly**
[be, cyrillic]

Noise applied: vowel 'U' > 'Yu', consonant 'r' > 'l', transposition at 3, duplication 'u'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Yuroychshcha Shnuuly	be	cyrillic
1	0.9815	Aubigny-sur-Nere	es	latin
2	0.9800	Juzjnouralsk	nb	latin
3	0.9796	Juzjnouralsk	nn	latin
4	0.9796	Juzjnouralsk	no	latin
5	0.9794	Rejon jezioroski	pl	latin
6	0.9787	Duniere-sur-Eyrieux	hr	latin
7	0.9785	Kurianki Pierwsze	tr	latin
8	0.9782	Aubigny-sur-Nere	id	latin
9	0.9780	Majuro District	und	latin
10	0.9776	Raionul Kirovske	ro	latin
11	0.9772	Aubigny-sur-Nere	es	latin
12	0.9771	Ostriw Kirowske	de	latin
13	0.9763	Cour-Saint-Maurice	es	latin
14	0.9762	Kurianki Pierwsze	pl	latin
15	0.9761	raion de Bureiski	ca	latin

Table 18: Noise robustness: **Lopatnitsy** $\rightarrow$ **Lpatnitsy** [und, cyrillic]

Noise applied: deletion 'o'					
Recovery: original not in top-k					
Rank	Score	Name	Lang	Script	
Q	—	Lpatnitsy	und	cyrillic	
1	0.9875	Colpon-Ata	it	latin	
2	0.9865	Tumpilima	ceb	latin	
3	0.9863	Tumpilima	und	latin	
4	0.9860	Gouimpleu I	und	latin	
5	0.9857	Colpon-Ata	tr	latin	
6	0.9840	Malnombre	en	latin	
7	0.9826	Malnombre	nl	latin	
8	0.9826	Sraindabey	en	latin	
9	0.9826	Sraindabey	en	latin	
10	0.9826	Sraindabey	en	latin	
11	0.9825	Cholpon-Ata	it	latin	
12	0.9805	Grindbach	und	latin	
13	0.9805	Temple Sound	sv	latin	
14	0.9803	Subalbandi	ast	latin	
15	0.9802	Colombier	sq	latin	

Table 19: Noise robustness: **Arri** $\rightarrow$ **AArri** [uk, cyrillic]

Noise applied: duplication 'A'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	AArri	uk	cyrillic
1	0.9972	Warri	fr	latin
2	0.9972	Marri	fr	latin
3	0.9972	Zarri	fr	latin
4	0.9972	Karri	fr	latin
5	0.9971	Qarri	fr	latin
6	0.9971	Harri	fr	latin
7	0.9970	Barri	fr	latin
8	0.9956	Raray	en	latin
9	0.9956	Raray	en	latin
10	0.9954	Raray	en	latin
11	0.9954	Saray	en	latin
12	0.9954	Karay	en	latin
13	0.9954	Naray	en	latin
14	0.9954	Faray	en	latin
15	0.9954	Taray	en	latin

Table 20: Noise robustness: **Golgat-Baltiston** $\rightarrow$ **Kolgat-Byatlistn** [tg, cyrillic]

Noise applied: deletion 'o', transposition at 9, vowel 'a' $\rightarrow$ 'ya', consonant 'G' $\rightarrow$ 'K'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Kolgat-Byatlistn	tg	cyrillic
1	0.9833	Recanto dos Eucaliptos	pt	latin
2	0.9798	castello di Cogoleto	it	latin
3	0.9776	caseta de Pepet lo Queto	ca	latin
4	0.9768	calle del Noviciado	sv	latin
5	0.9762	stazione di Regalbuto	it	latin
6	0.9762	castillo de Cogolludo	es	latin
7	0.9761	calle de Isabel Tintero	es	latin
8	0.9760	castle of Roccatagliata	en	latin
9	0.9760	budynek d. klasztoru	pl	latin
10	0.9760	Skollersta prastang	ro	latin
11	0.9759	Estancia Eucaliptus	und	latin
12	0.9757	case de sante tall bakhle	fr	latin
13	0.9757	castello di Pergolato	it	latin
14	0.9756	Alto Del Eucalipto	es	latin
15	0.9753	Estacao Eucaliptos	pt	latin

Table 21: Noise robustness: **Melekesskiy Rayon** → **Meleeskiy Ryon** [ru, cyrillic]

Noise applied: deletion 's', deletion 'k', deletion 'a'					
Recovery: original not in top-k					
Rank	Score	Name	Lang	Script	
Q	—	Meleeskiy Ryon	ru	cyrillic	
1	0.9899	Balogun Market	en	latin	
2	0.9875	Sar Band-e Owkar	und	latin	
3	0.9874	Block Twenty	und	latin	
4	0.9865	Ballykevan West	en	latin	
5	0.9860	Black Boney Bend	en	latin	
6	0.9858	Sally Rocks, Lobulo	und	latin	
7	0.9851	Ballykevan East	en	latin	
8	0.9850	Bluegum Park	ceb	latin	
9	0.9850	Bluegum Park	und	latin	
10	0.9839	Black Sound Point	en	latin	
11	0.9838	Zalakoveskut	und	latin	
12	0.9838	Zalakoveskut	eu	latin	
13	0.9837	Sham Baluk	und	latin	
14	0.9837	Zalakoveskut	ast	latin	
15	0.9837	Zalakoveskut	eo	latin	

Table 22: Noise robustness: **Sent-Antuan-d’ObegHosh** $\rightarrow$ **Sent-Antuan-’ObkheHsho** [ce, cyrillic]

Noise applied: deletion ’d’, transposition at 18, transposition at 15, consonant ’g’>’kh’
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Sent-Antuan-’ObkheHsho	ce	cyrillic
1	0.9714	sentier Autour de Cognac	fr	latin
2	0.9675	cenotaph of Jean-Jacques Rousseau	en	latin
3	0.9643	poste de sante Sinthioumogo	fr	latin
4	0.9639	synagogue de Reichenberg	fr	latin
5	0.9598	saint-etienne in jargeau	nl	latin
6	0.9584	synagogue de Saint-Ingbert	fr	latin
7	0.9577	synagoge van Augsburg	nl	latin
8	0.9548	zricenina hradu Quingburk	fr	latin
9	0.9535	puente colgante de Conwy	es	latin
10	0.9531	kantono Sankt-Jozefo	eo	latin
11	0.9526	synagogue d’Aschaffembourg	fr	latin
12	0.9524	sente du Vieux Pont de Bois	mul	latin
13	0.9519	sonnenhotel birkenhof	und	latin
14	0.9487	sanctuaire faunique de Bor	fr	latin
15	0.9482	sao benedito-Ceara	en	latin

Table 23: Noise robustness: **Ulitsa Obilicev venats** → **Ullits Obilicevv vema**
[sr, cyrillic]

Noise applied: duplication 'v', consonant 'n' > 'm', deletion 'a', duplication 'l', deletion 'ts', transposition at 13
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Ullits Obilicevv vema	sr	cyrillic
1	0.9891	O'Keefe Reservoir Number Six	und	latin
2	0.9889	North Branch Rapide De Femme	und	latin
3	0.9888	Beach Grove Church Number 2	und	latin
4	0.9884	Beach Grove Church Number 1	und	latin
5	0.9883	New Dresser Number 4 Dam	ceb	latin
6	0.9883	New Dresser Number 4 Dam	und	latin
7	0.9883	Poblacion Obispo Camus	und	latin
8	0.9881	Bawan fata musuem Calabar	ha	latin
9	0.9876	Mazra'eh-ye Abranabad	und	latin
10	0.9875	Powell Prospect Number One	en	latin
11	0.9875	Cold River Forest Camp	und	latin
12	0.9875	Ab Anbar-e Shah 'Abbasi	und	latin
13	0.9874	Willis River Number Six Dam	ceb	latin
14	0.9874	Eperjes-hegyi 1.sz. barlang	und	latin
15	0.9873	Beech Grove Church Number 2	und	latin

Table 24: Noise robustness: **Tzami toy Zarkent** $\rightarrow$ **Tsan toy Zarrkent** [el, greek]

Noise applied: consonant 'm'>'n', deletion 'i', consonant 'z'>'s', duplication 'r'					
Recovery: original not in top-k					
Rank	Score	Name	Lang	Script	
Q	—	Tsan toy Zarrkent	el	greek	
1	0.9878	queens jubilee 2012 wood	und	latin	
2	0.9838	cite Saint-Georges	en	latin	
3	0.9795	changi hospital	gan	latin	
4	0.9768	canton d'Issigeac	ga	latin	
5	0.9762	canton d'Issigeac	sl	latin	
6	0.9762	canton d'Issigeac	vec	latin	
7	0.9754	chiese e conventi di Goa	it	latin	
8	0.9743	sanctuaire Myogi	fr	latin	
9	0.9729	Szentesica-forras	und	latin	
10	0.9724	jinganshushanxibefeng	zh	latin	
11	0.9714	szentesi tevetorony	hu	latin	
12	0.9707	quai Saint-Exupery	en	latin	
13	0.9698	sentier des Theux	mul	latin	
14	0.9691	tombe de saint Pierre	fr	latin	
15	0.9685	canyon des Aiguieres	fr	latin	

Table 25: Noise robustness: **Agia Thekla** $\rightarrow$ **Aga ThThThekla** [und, greek]

Noise applied: deletion 'i', duplication 'Th', duplication 'Th'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Aga ThThThekla	und	greek
1	0.9874	Paktal-chae	sv	latin
2	0.9874	Nachtegal	ast	latin
3	0.9871	Nachtegal	und	latin
4	0.9837	Vettakkal	nl	latin
5	0.9835	SD Wakatitir	en	latin
6	0.9832	Paktalgoji	und	latin
7	0.9831	Pattadakal	nl	latin
8	0.9830	Aghateggal	und	latin
9	0.9828	Heeghtakker	nl	latin
10	0.9823	Paktalchae	und	latin
11	0.9815	Eegatuak Hill	en	latin
12	0.9815	Chulkati Gang	en	latin
13	0.9815	Nighthawk Hill	en	latin
14	0.9810	Eegatuak Hill	nl	latin
15	0.9809	Meghchatar	en	latin

Table 26: Noise robustness: **Mpongksangk-ri** → **Mpomngskangk-ri** [el, greek]

Noise applied: transposition at 5, duplication 'n', consonant 'n'>'m'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Mpomngskangk-ri	el	greek
1	0.9779	Kampung Nakongwe	und	latin
2	0.9763	Dongmacunhecun	en	latin
3	0.9758	Sanjoinokumacho	en	latin
4	0.9747	Donggo-mokjang	und	latin
5	0.9743	Puukonmaankangas	und	latin
6	0.9737	Songgang-mokjang	und	latin
7	0.9725	Hna hkaung River	my	latin
8	0.9724	Lungqungarmo La	und	latin
9	0.9713	Bokangaon Tanda	und	latin
10	0.9713	Hna Hkaung Pauk	my	latin
11	0.9712	Lungqungarmo La	bo	latin
12	0.9712	Lungqungarmo La	ceb	latin
13	0.9710	Sammakkonotko	und	latin
14	0.9708	Mangkono Seamount	und	latin
15	0.9699	Kampong Sangkor	und	latin

Table 27: Noise robustness: **Velies** $\rightarrow$ **eVlies** [el, greek]

Noise applied: transposition at 0
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	eVlies	el	greek
1	0.9932	Avles	und	latin
2	0.9932	Avlos	nb	latin
3	0.9929	Avlos	no	latin
4	0.9929	Avlos	und	latin
5	0.9928	Avlos	nn	latin
6	0.9924	Avles	ga	latin
7	0.9906	Bulise	en	latin
8	0.9904	Mulise	en	latin
9	0.9901	Mollasi	hu	latin
10	0.9897	Devlez	und	latin
11	0.9897	Mollasi	sq	latin
12	0.9894	Molise	en	latin
13	0.9894	Molise	en	latin
14	0.9892	Haulies	ms	latin
15	0.9890	Dawleys	en	latin

Table 28: Noise robustness: **Dimos diamerismatos Alitoys** → **Dimos di-amerismatos Alitoys** [el, greek]

Noise applied: vowel 'o' > 'o'

Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Dimos diamerismatos Alitoys	el	greek
1	0.9953	Parque internacional La Amistad	es	latin
2	0.9926	Parque Internacional La Amistad	es	latin
3	0.9923	Diocese de Hamilton nas Bermudas	pt	latin
4	0.9923	Diocese de Hamilton nas Bermudas	pt	latin
5	0.9918	Kriz v Dobrem mistni cast Chmeliste	cs	latin
6	0.9914	MTs Raudhotul Hasanah Al-Muhammady	id	latin
7	0.9912	Ev.-Luth. Kirchengemeinde Bad Bramstedt	de	latin
8	0.9912	Smiths Farmhouse and attached walls	en	latin
9	0.9903	Ehemaliges Bauernhaus und Schmiede	de	latin
10	0.9901	Teesta Agro Industries Limited	und	latin
11	0.9899	stade municipal de Saint-Cyr-en-Arthies	fr	latin
12	0.9898	Nebengebäude Hofstrasse 8 in Himmelstadt	en	latin
13	0.9892	Umspannanlage Darmstadt-Nord	de	latin
14	0.9891	Round cairn 557m SSW of Caradon Hill summit	en	latin
15	0.9888	centrale thermique Kashima Kyodo	fr	latin

Table 29: Noise robustness: **Aithikia** $\rightarrow$ **Aithykiaa** [grc, greek]

Noise applied: duplication 'a', vowel 'i'>'y'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Aithykiaa	grc	greek
1	0.9935	Achgueig	und	latin
2	0.9925	Ityaikya	und	latin
3	0.9897	Fajticka	und	latin
4	0.9892	Uji-gawa	ja	latin
5	0.9886	Sekikasi	tr	latin
6	0.9886	Dashoji-gawa	ja	latin
7	0.9885	Utoykaia	pcd	latin
8	0.9885	Utoykaia	la	latin
9	0.9885	Utoykaia	wa	latin
10	0.9885	Aegi-gogae	sv	latin
11	0.9882	Aegi-gogae	ceb	latin
12	0.9882	Utoykaia	li	latin
13	0.9882	Utoykaia	nb	latin
14	0.9878	Utoykaia	vls	latin
15	0.9872	Lajatico	it	latin

Table 30: Noise robustness: **Mprienon-syr-Armanson** $\rightarrow$ **Mprienon-sy-Armaamsonn** [el, greek]

Noise applied: duplication 'n', duplication 'a', deletion 'r', consonant 'n'>'m'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Mprienon-sy-Armaamsonn	el	greek
1	0.9778	Norra Hamnbassangen	und	latin
2	0.9749	kanton Brienon-sur-Armancon	nl	latin
3	0.9749	Hermann Grunspan	und	latin
4	0.9747	Tamuning-Tumon-Harmon Mayor's Office	und	latin
5	0.9744	Alqueria de Los Pampanos	und	latin
6	0.9743	Kust skvazhin No700 Alkinskogo neftyanogo mestorozhdeniya	und	cyrillic
7	0.9735	Krasnopresnenskaya Embankment	en	latin
8	0.9734	Termas romanas de Acinipo	es	latin
9	0.9731	Colonia Lomas del Camino	es	latin
10	0.9728	Zyamlyanka shtaba 1-y Bykhauskay partyzanskay brygady	und	cyrillic
11	0.9724	Zyamlyanka shtaba 1-y Bykhauskay partyzanskay brygady	be	cyrillic
12	0.9723	ryoma no umareta machi kinenkan	fr	latin
13	0.9722	Offenlandbereiche um Grimminghausen	en	latin
14	0.9722	Offenlandbereiche um Grimminghausen	en	latin
15	0.9722	Offenlandbereiche um Grimminghausen	en	latin

Table 31: Noise robustness: **Kamizano** $\rightarrow$ **Komizaon** [el, greek]

Noise applied: vowel 'a'>'o', transposition at 6
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Komizaon	el	greek
1	0.9684	kwamchina	und	latin
2	0.9642	koumassan	und	latin
3	0.9614	commusPoMo	und	latin
4	0.9534	cour Vincent	fr	latin
5	0.9469	Eikemofoss	und	latin
6	0.9469	koce mis	tr	latin
7	0.9464	kokoomus	fi	latin
8	0.9463	quemao si	en	latin
9	0.9440	comitium	it	latin
10	0.9425	kougommuur	af	latin
11	0.9415	kemasan	und	latin
12	0.9414	Akoumassi	und	latin
13	0.9412	Skavemose	und	latin
14	0.9412	jimingshan	zh	latin
15	0.9384	Makivehmas	und	latin

Table 32: Noise robustness: **Palloyriotissa** → **Palroiotisssa** [en, greek]

Noise applied: deletion 'r', deletion 'y', consonant 'l'>'r', duplication 's'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	Palroiotisssa	en	greek
1	0.9715	place Milosz	fr	latin
2	0.9683	place Travot	en	latin
3	0.9679	palazzo Poeti	en	latin
4	0.9679	parc loisirs	fr	latin
5	0.9678	palais Grossi	fr	latin
6	0.9677	lower park wood	en	latin
7	0.9674	palais Curtius	fr	latin
8	0.9671	poplar house	en	latin
9	0.9650	place Valois	fr	latin
10	0.9647	plaza ferias	und	latin
11	0.9642	place Carros	fr	latin
12	0.9639	palais Troise	fr	latin
13	0.9632	palus Boebeis	en	latin
14	0.9631	place Charost	fr	latin
15	0.9629	palais Krausz	fr	latin

Table 33: Noise robustness: **Seli Artas** $\rightarrow$ **eli Artas** [el, greek]

Noise applied: deletion 'S'

Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	eli Artas	el	greek
1	0.9902	Pal Seratus	ms	latin
2	0.9880	Tel Irit	und	latin
3	0.9864	Lehriabad	und	latin
4	0.9861	Mala Varda	en	latin
5	0.9860	El Aradat	und	latin
6	0.9858	Meharabad	und	latin
7	0.9857	Beharabad	ku	latin
8	0.9857	El Fritis	en	latin
9	0.9856	Zla Greda	en	latin
10	0.9856	Nelle Barat	es	latin
11	0.9854	Ali-ret	und	latin
12	0.9852	Jelec Grad	sr	latin
13	0.9850	Terra Alta	uz	latin
14	0.9850	Alle Crede	pt	latin
15	0.9849	Val-Paradis	en	latin

Table 34: Noise robustness: **pi’nryj rynch hwi’y d** $\rightarrow$ **ppi’mrryj rych hwi’y d**
[ur, arabic]

Noise applied: duplication 'r', consonant 'n'>'m', transposition at 11, deletion 'n', duplication 'p'				
Recovery: original not in top-k				
Rank	Score	Name	Lang	Script
Q	—	ppi’mrryj rych hwi’y d	ur	arabic
1	0.9882	Zembrzus-Mokry Grunt	tr	latin
2	0.9878	Coombes Croft Library	en	latin
3	0.9878	Coombes Croft Library	en	latin
4	0.9878	Coombes Croft Library	en	latin
5	0.9875	Quebrada Apanacusha	und	latin
6	0.9871	provincia di Chumbivilcas	it	latin
7	0.9862	Bembesi Forest Area	und	latin
8	0.9849	Operacija Erntefest	sh	latin
9	0.9846	Samara passenger port	en	latin
10	0.9841	Rue Ambroise Croizat	tr	latin
11	0.9839	Noyabrsk Airport	ceb	latin
12	0.9839	Noyabrsk Airport	und	latin
13	0.9835	Bumbeyskiy Khrebet	en	latin
14	0.9833	Menara Emirates Office	id	latin
15	0.9830	Gempa bumi Tabriz 1641	id	latin

Table 35: Noise robustness: **kwh myshb** $\rightarrow$ **kwh myshp** [fa, arabic]

Noise applied: consonant 'b' > 'p'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	kwh myshp	fa	arabic
1	0.9760	Simacota	ga	latin
2	0.9753	Namikata	es	latin
3	0.9752	Simacota	avk	latin
4	0.9751	Limacota	und	latin
5	0.9750	Simacota	ast	latin
6	0.9750	Simacota	cy	latin
7	0.9750	Simacota	gl	latin
8	0.9750	Simacota	war	latin
9	0.9750	Oimacota	und	latin
10	0.9750	Kamikita	vi	latin
11	0.9750	Simacota	ceb	latin
12	0.9750	Simacota	nan	latin
13	0.9750	Simacota	und	latin
14	0.9749	Chhamkot	und	latin
15	0.9730	Hamaquita	es	latin

Table 36: Noise robustness: **dyrwng bkwng** $\rightarrow$ **dyrwng bkwng** [ms, arabic]

Noise applied: transposition at 6
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	dyrwng bkwng	ms	arabic
1	0.9847	Tampojung Guwa	nl	latin
2	0.9835	Mokonokpor	und	latin
3	0.9824	Bungurcopong	id	latin
4	0.9803	Shongopovi	und	latin
5	0.9801	Cukangbungur	id	latin
6	0.9801	Shongopovi	eu	latin
7	0.9801	Shongopovi	gl	latin
8	0.9799	Shongopovi	ceb	latin
9	0.9799	Shongopovi	ga	latin
10	0.9799	Shongopovi	lld	latin
11	0.9799	Shongopovi	sh	latin
12	0.9799	Shongopovi	vo	latin
13	0.9798	Shongopovi	hbs	latin
14	0.9798	Shongopovi	nan	latin
15	0.9793	Bungurcopong	ms	latin

Table 37: Noise robustness: **spylkww** $\rightarrow$ **spylkw** [fa, arabic]

Noise applied: vowel 'w'>"

Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	spylkw	fa	arabic
1	0.9769	sajbaki	tr	latin
2	0.9609	sga yab	bo	latin
3	0.9502	sheep geo	en	latin
4	0.9442	Fischelbach	fi	latin
5	0.9423	sybieki	tr	latin
6	0.9420	shekhpura	en	latin
7	0.9407	2024Nian9YueDangLao-TeLangPuYuCiAn	zh	cjk
8	0.9406	shipo he	zh	latin
9	0.9393	soba 828	sl	latin
10	0.9390	2019Nian1YueLinKenJiNianTangDuiZhiShiJian	zh	cjk
11	0.9387	shifu he	zh	latin
12	0.9384	2023Nian9YuePingDongKeJiChanYeYuanQuHuoZai	zh	cjk
13	0.9384	shibahu	und	latin
14	0.9383	Essehb	ceb	latin
15	0.9383	2023Nian9YuePingDongKeJiChanYeYuanQuHuoZai	zh	cjk

Table 38: Noise robustness: **kwh sr gdr syf ldyn** → **kwh sr gdr syff yldyn** [fa, arabic]

Noise applied: vowel " >'y', duplication 'f', consonant 'd' >'d'					
Recovery: original not in top-k					
Rank	Score	Name	Lang	Script	
Q	—	kwh sr gdr syff yldyn	fa	arabic	
1	0.9818	Vieux Pont de Belcastel	nl	latin	
2	0.9809	Majada los Tanquitos	es	latin	
3	0.9806	Minas del Marquesado	ceb	latin	
4	0.9797	Escola Elisa dos Santos	pt	latin	
5	0.9789	Mesa Mezquitalito	en	latin	
6	0.9787	Estancia El Pescado	und	latin	
7	0.9787	Finca Cana Castilla	es	latin	
8	0.9783	Las Banquetas del Ojito	es	latin	
9	0.9780	Ragaciems lighthouse	en	latin	
10	0.9773	St. Nickolas Hospital	en	latin	
11	0.9766	El Rosalito Escondido	es	latin	
12	0.9764	Wescott Oil Field	en	latin	
13	0.9762	Vetle Skagastolstind	es	latin	
14	0.9759	Vetle Skagastolstind	en	latin	
15	0.9757	Pig Styes at Glascoed Mill	en	latin	

Table 39: Noise robustness: **sd qtrm** $\rightarrow$ **d qtrm** [und, arabic]

Noise applied: deletion 's', transposition at 3
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	d qtrm	und	arabic
1	0.9872	Vikebygd	ms	latin
2	0.9854	Vikebygd	en	latin
3	0.9837	Okpagode	und	latin
4	0.9822	Aguatita	und	latin
5	0.9821	Aguatita	sh	latin
6	0.9820	Aguatita	ga	latin
7	0.9811	Aguadita	und	latin
8	0.9803	Hkawbude	und	latin
9	0.9802	Akoubaida	en	latin
10	0.9796	Vikebygd	nl	latin
11	0.9785	Matakidi	und	latin
12	0.9778	Pataquid	und	latin
13	0.9778	Okuabude	und	latin
14	0.9777	Adikade	ga	latin
15	0.9776	Pataquid	fil	latin

Table 40: Noise robustness: **lyk si'd, whi'yw** → **lyk thwd, wwwhy'w** [ur, arabic]

Noise applied: vowel " >'w", deletion 'i", transposition at 13, vowel " >'w", consonant 's' >'th'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	lyk thwd, wwwhy'w	ur	arabic
1	0.9710	la Costeta	ca	latin
2	0.9691	lac Testard	it	latin
3	0.9688	lightwood	en	latin
4	0.9682	Tanquecitos	es	latin
5	0.9676	lodgeside	en	latin
6	0.9654	lydgates	en	latin
7	0.9640	BCSicletos	es	latin
8	0.9632	Muncheroda	it	latin
9	0.9630	les Cotetes	ca	latin
10	0.9620	Minatosato	en	latin
11	0.9619	Pino Hachado	es	latin
12	0.9617	l'Institut	ca	latin
13	0.9599	Vauclusotte	fr	latin
14	0.9598	lac Testard	de	latin
15	0.9596	Cocklet Wood	en	latin

Table 41: Noise robustness: **mk nwrt** $\rightarrow$ **k nwrtt** [fa, arabic]

Noise applied: deletion 'm', duplication 't'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	k nwrtt	fa	arabic
1	0.9721	Pommerot	und	latin
2	0.9709	unmarked	und	latin
3	0.9672	Dom emeryta	pl	latin
4	0.9668	Gummeroth	ms	latin
5	0.9664	Gummeroth	fi	latin
6	0.9659	Ommeroth	id	latin
7	0.9657	Kimmirut	it	latin
8	0.9655	Ommeroth	fi	latin
9	0.9642	Gummgrat	de	latin
10	0.9639	Ommeroth	ms	latin
11	0.9637	Kommherut	de	latin
12	0.9633	Gummeroth	vi	latin
13	0.9622	Rommerode	vi	latin
14	0.9619	Cammarota	ro	latin
15	0.9617	Ommeroth	sw	latin

Table 42: Noise robustness: **mylbrwk** $\rightarrow$ **mlybrwkk** [ur, arabic]

Noise applied: transposition at 1, duplication 'k'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	mlybrwkk	ur	arabic
1	0.9754	Ambellako	sv	latin
2	0.9752	Smalabrekka	no	latin
3	0.9747	Smalabrekka	ceb	latin
4	0.9747	Smalabrekka	und	latin
5	0.9744	Ambellako	en	latin
6	0.9740	Ambelakia	en	latin
7	0.9726	Embeleco	en	latin
8	0.9723	Inabarraga	und	latin
9	0.9710	Kumlubucak	ca	latin
10	0.9707	Paalbeekweg	nl	latin
11	0.9703	Gambler Lake	fr	latin
12	0.9702	Ambalapaka	und	latin
13	0.9699	Kumlubucak	es	latin
14	0.9698	Ambler Park	fr	latin
15	0.9698	Malbarqouq	nl	latin

Table 43: Noise robustness: **rtf't gw bst** $\rightarrow$ **rtf't gw bstt** [fa, arabic]

Noise applied: transposition at 4, duplication 't'
Recovery: original not in top-k

Rank	Score	Name	Lang	Script
Q	—	rtf't gw bstt	fa	arabic
1	0.9931	Skerry of Wastbist	en	latin
2	0.9920	Cape Sharp Lighthouse	en	latin
3	0.9909	barrage des Eaux-Mortes	fr	latin
4	0.9907	Cerco de Budapeste	pt	latin
5	0.9900	Prostastigbukta	no	latin
6	0.9900	Prostastigbukta	und	latin
7	0.9898	Rocket Apparatus House	en	latin
8	0.9897	Hartig bei Hochstadt	es	latin
9	0.9893	parrocchia di Webster	it	latin
10	0.9891	Lake Superior Estates	en	latin
11	0.9890	baza ot dyha Gorec	und	latin
12	0.9888	gare de Rorschach Stadt	fr	latin
13	0.9878	Marie Stopes Society	en	latin
14	0.9877	Rubha Aird a' Chothais	gd	latin
15	0.9877	Step' Darvazgaydasht	ru	latin

Table 44: Noise robustness evaluation summary

Metric	Value
Total queries	40
Original recovered in top-k	0 (0.0%)
Original at rank 1	0 (0.0%)
Original in top-5	0 (0.0%)
Original in top-10	0 (0.0%)
Mean recovery rank	inf
Mean top-1 similarity	0.9838

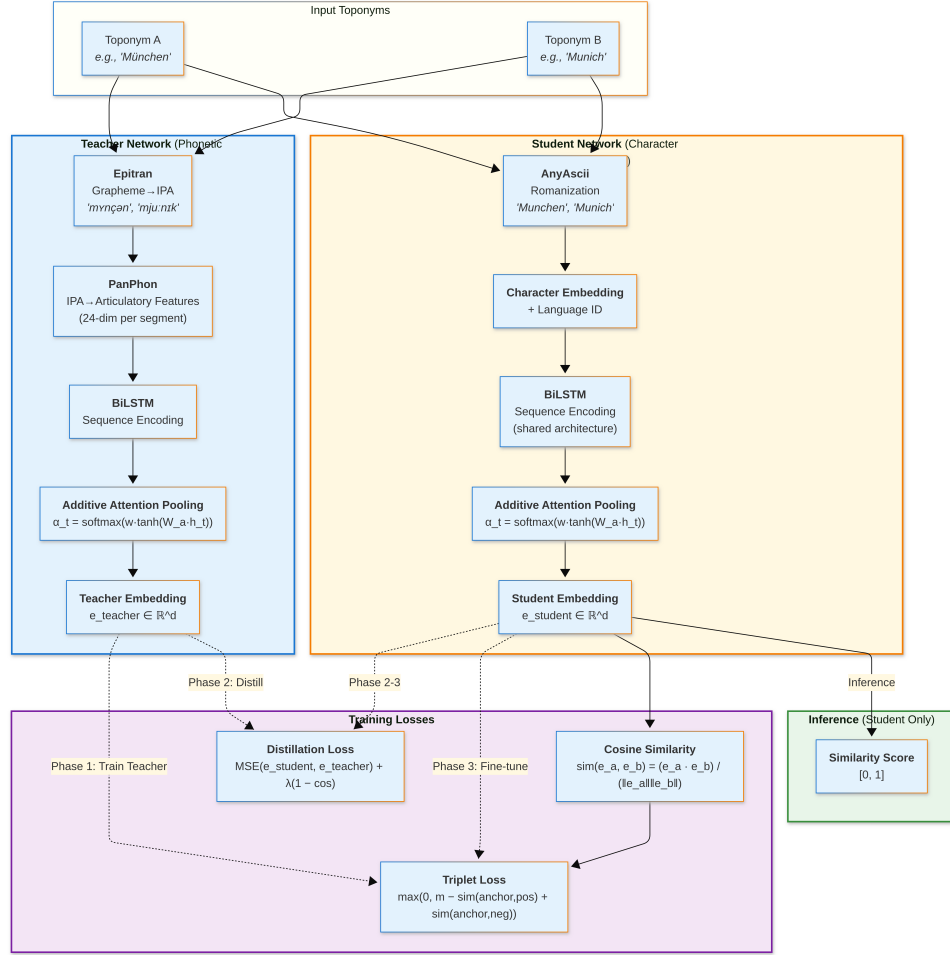


Figure 1: Student-Teacher architecture for phonetic similarity learning. The Teacher network processes IPA articulatory features via Epitrans and PanPhon; the Student network processes romanized characters with language embeddings. Both use BiLSTM encoding followed by additive attention pooling. During training, the Student learns to approximate Teacher embeddings through knowledge distillation (Phases 1–3). At inference, only the Student network is required, outputting cosine similarity scores.

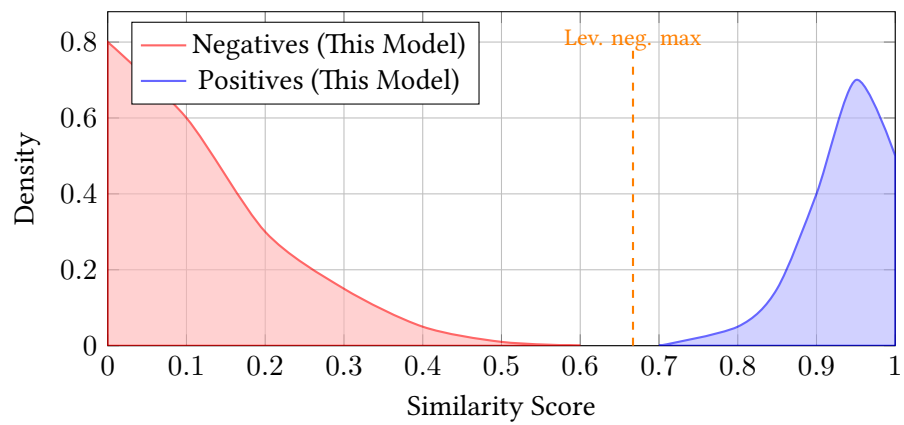


Figure 2: Similarity score distributions for positive and negative pairs. The trained model concentrates negative pairs near zero (95th percentile = 0.485), creating a wide margin for threshold selection. Levenshtein negatives extend to 0.667, reducing the viable operating range.